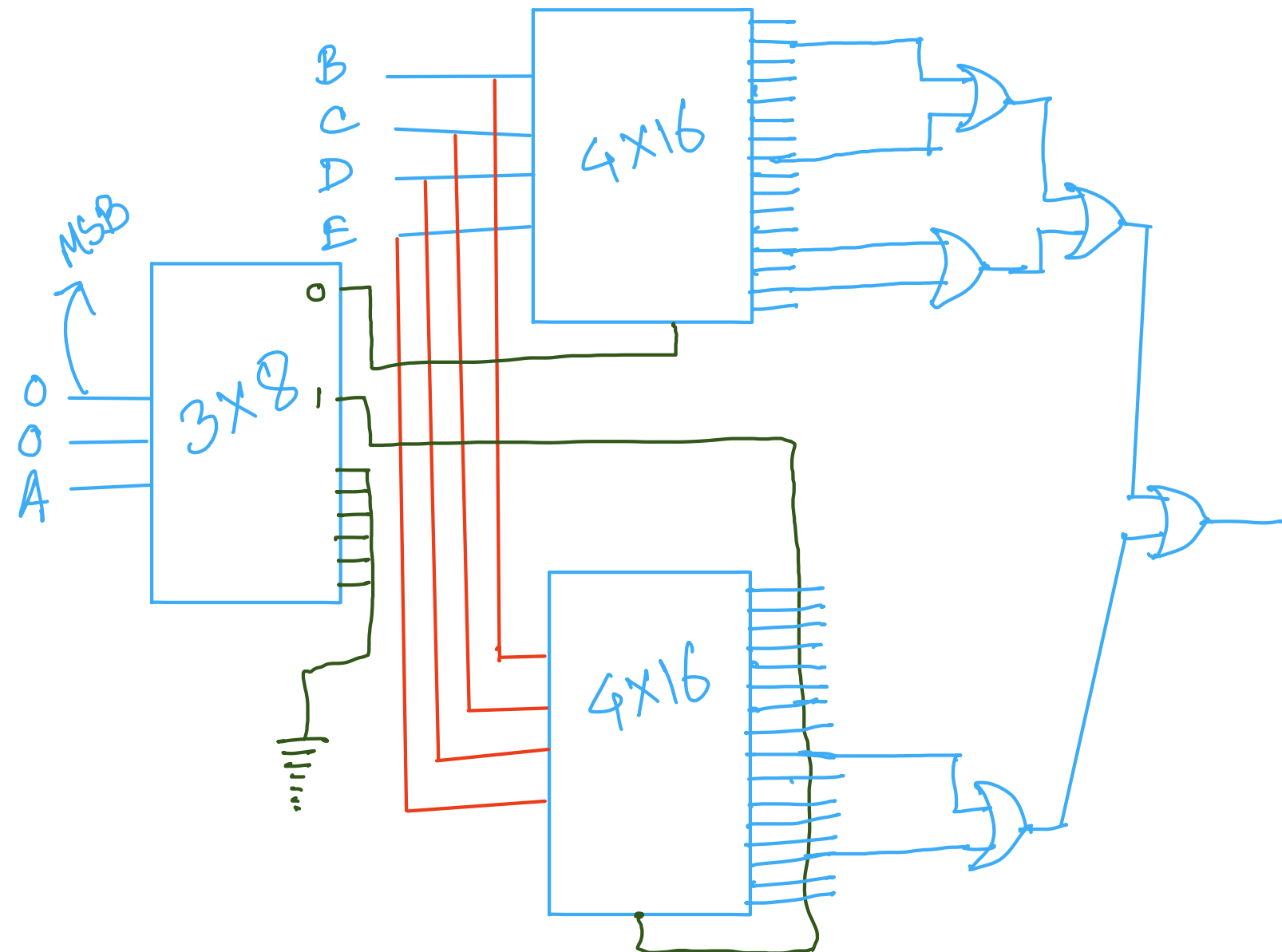


For All questions, multiple solutions are available. I showed only one possible way.

1.CO3	Build the following function using both 4x16 decoder(s) and 3x8 decoder(s) in a single circuit: $F(A,B,C,D,E) = \sum (1, 7, 12, 14, 24, 30)$ In your circuit, the number of 4x16 decoder(s) must be more than the number of 3x8 decoder(s).
2.CO3	Implement the boolean function using both 4:1 MUX(s) and 2:1 MUX(s) in a single circuit. $F(A,B,C,D) = \sum (2, 1, 4, 5, 9, 14, 15)$ In your circuit, the number of 4:1 mux(s) must be more than the number of 2:1 mux(s).

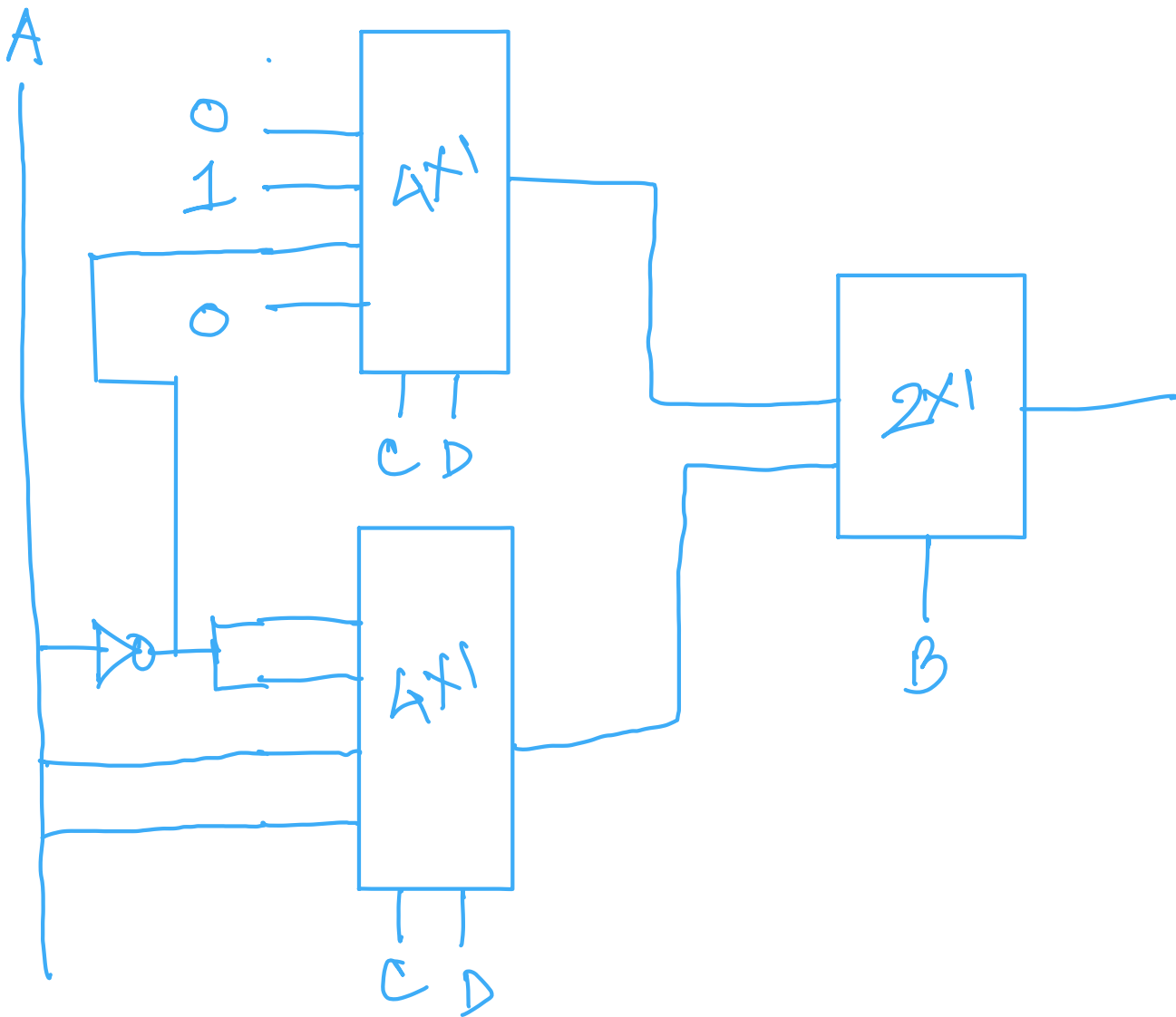
Solution

1

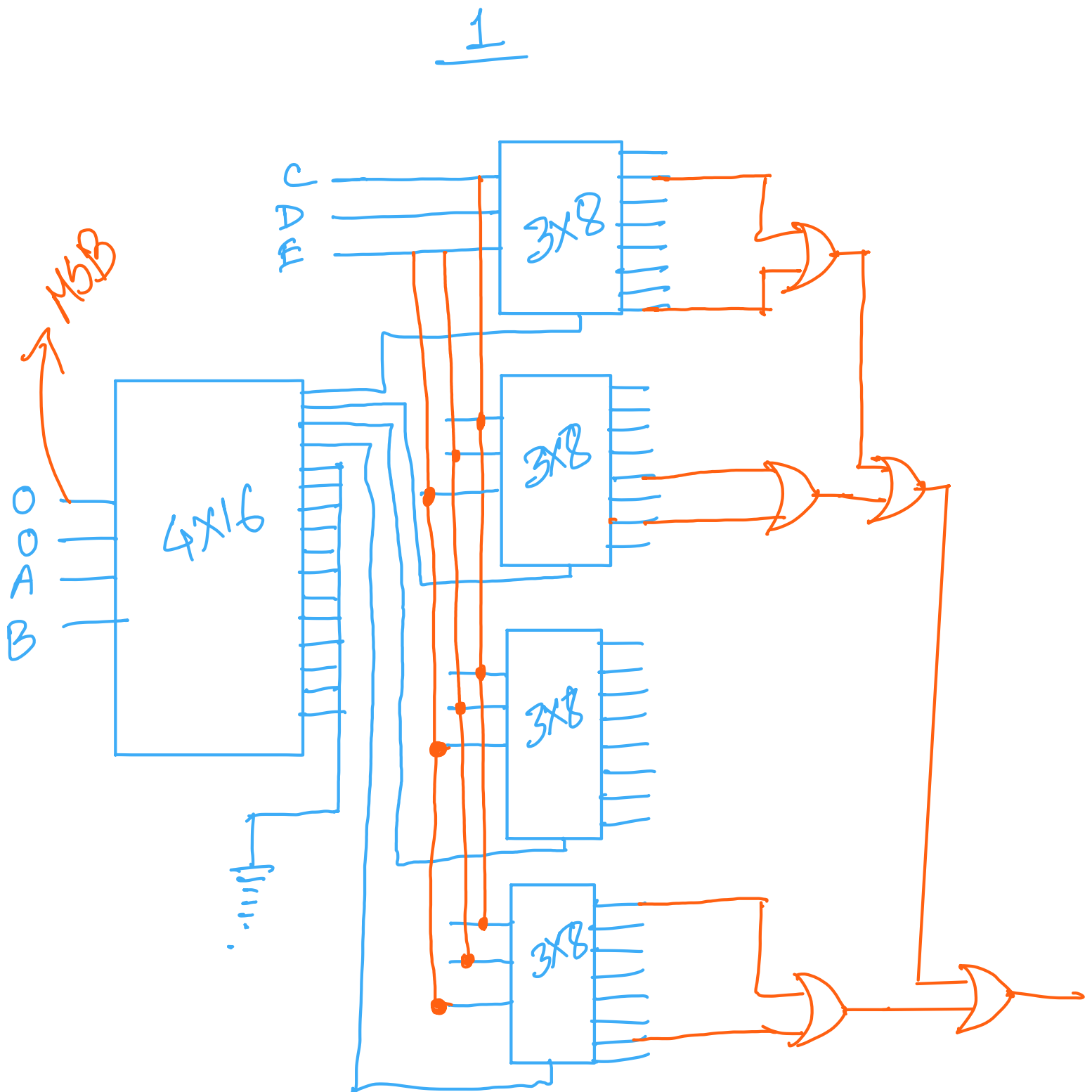


2

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
A'	0	①	②	3	④	⑤	6	7
A	8	⑨	10	11	12	13	⑭	⑮
	0	1	A'	0	A'	A'	A	A



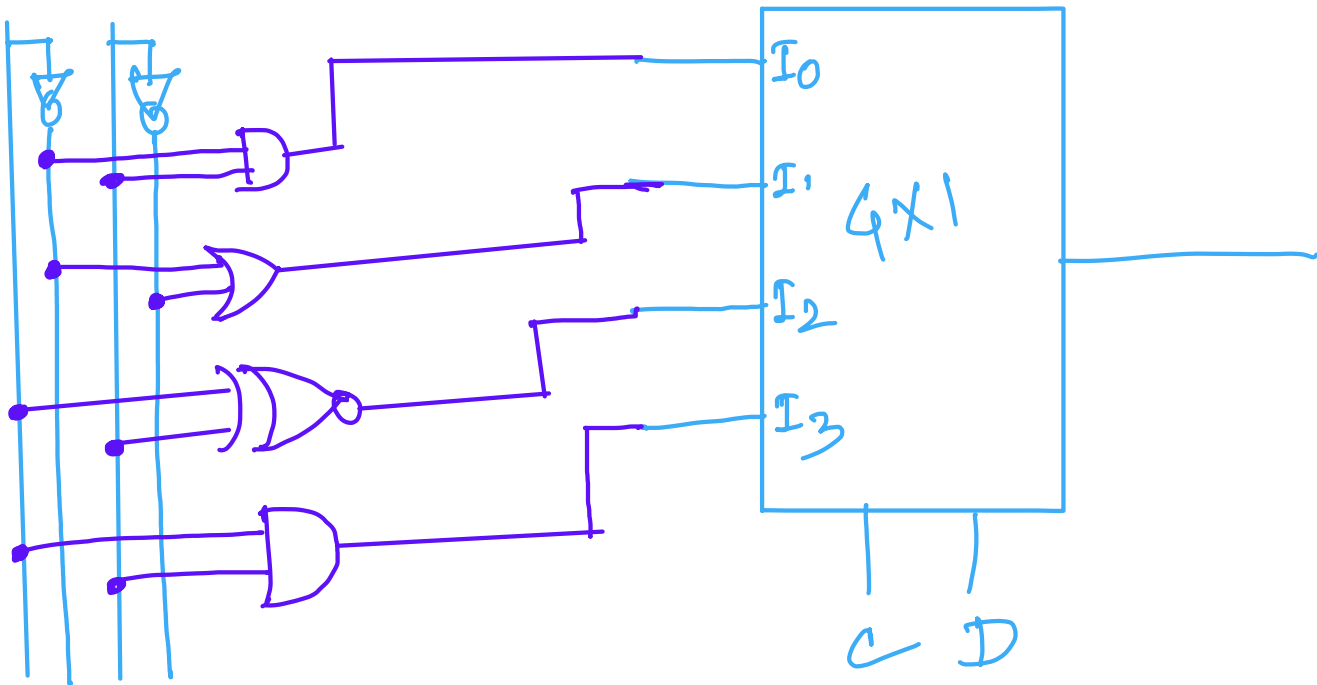
1.CO3	Build the following function using both 4x16 decoder(s) and 3x8 decoder(s) in a single circuit: $F(A,B,C,D,E) = \sum (1, 7, 12, 14, 24, 30)$ In your circuit, the number of 4x16 decoder(s) must be less than the number of 3x8 decoder(s).
2.CO3	Implement the boolean function using only one 4:1 MUX. $F(A,B,C,D) = \sum (2, 1, 4, 5, 9, 14, 15)$



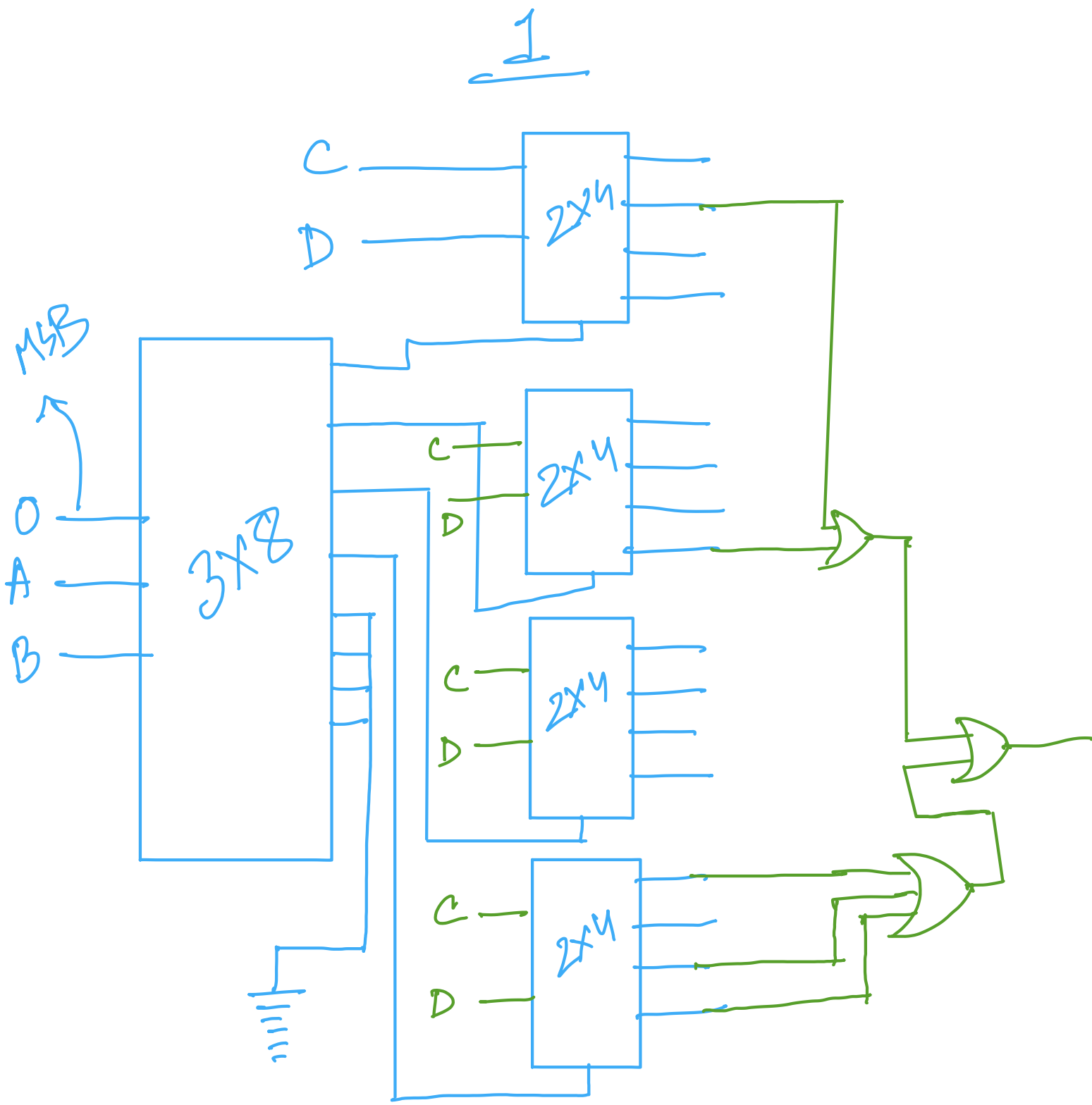
2

	I_0	I_1	I_2	I_3
$A'B'$	0	①	②	3
$A'B$	④	⑤	6	7
AB'	8	⑨	10	11
AB	12	13	⑭	⑮
	$A'B$	$A'+B'$	$A \odot B$	AB

A B



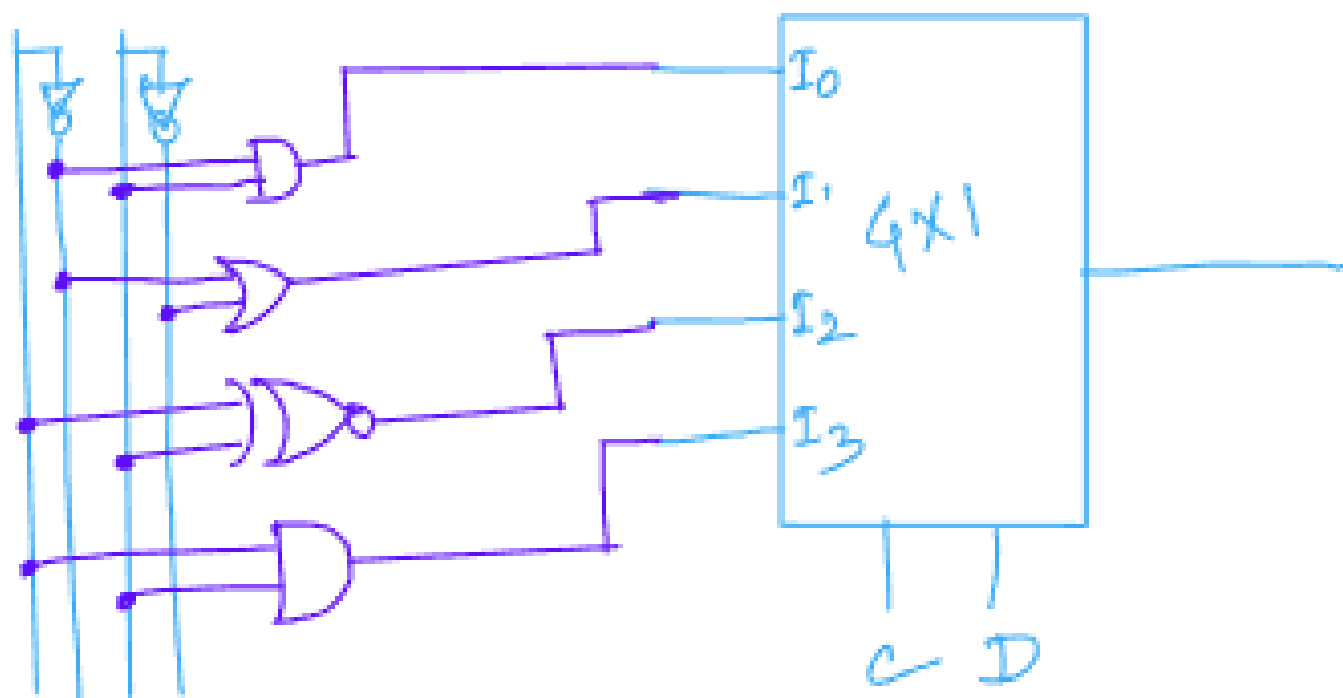
1.CO3	Build the following function using both 3x8 decoder(s) and 2x4 decoder(s) in a single circuit: $F(A,B,C,D) = \sum (1, 7, 12, 14, 15)$ In your circuit, the number of 3x8 decoder(s) must be less than the number of 2x4 decoder(s).
2.CO3	Implement the boolean function using only one 4:1 MUX. $F(A,B,C,D) = \sum (2, 1, 4, 5, 9, 14, 15)$



2

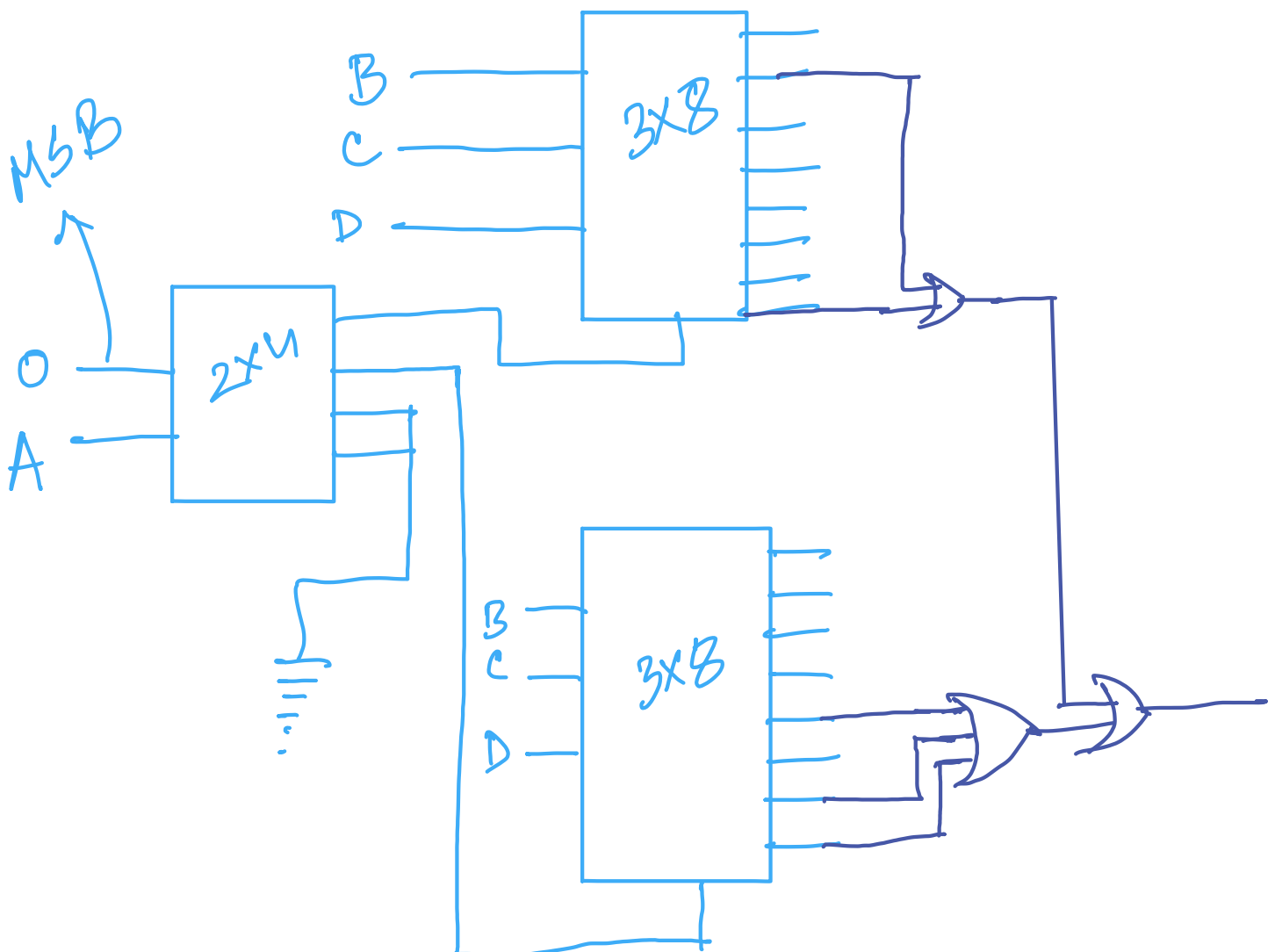
	I_0	I_1	I_2	I_3
$A'B'$	0	①	②	3
$A'B$	④	⑤	6	7
AB'	8	⑨	10	11
AB	12	13	⑭	⑮
	$A'B$	$A+B'$	$A \odot B$	AB

A B



1.CO3	Build the following function using both 3x8 decoder(s) and 2x4 decoder(s) in a single circuit: $F(A,B,C,D) = \sum (1, 7, 12, 14, 15)$ In your circuit, the number of 3x8 decoder(s) must be more than the number of 2x4 decoder(s).
2.CO3	Implement the boolean function using both 4:1 MUX(s) and 2:1 MUX(s) in a single circuit. $F(A,B,C,D) = \sum (2, 1, 4, 5, 9, 14, 15)$ In your circuit, the number of 4:1 mux(s) must be less than the number of 2:1 mux(s).

1



2

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
A'	0	①	②	3	④	⑤	6	7
A	8	⑨	10	11	12	13	⑭	⑮
	0	1	A'	0	A'	A'	A	A

